	Part – B (4 x 4 = 16 Marks) Instructions: Answer any 4					
11	Convert the following numbers as specified: a) Convert the hexadecimal number 2F9B to its binary and decimal equivalents. b) Convert the binary number 1101011.101 to its octal and hexadecimal equivalents.	4	2	1	2	2.4.1
12	Perform the following binary arithmetic operations using the 2's complement method: a) Subtract 1101 from 10001 and provide the result in binary. b) Add -7 (in 2's complement form) to 9 (in 2's complement form) and give the result in decimal form.	4	2	1	2	2.4.1
13	Design and explain a digital circuit using only NAND gates that function as an XOR gate. Provide the logic expression and draw the corresponding circuit diagram.	4	2	1	2	2.4.1
14	Differentiate between direct and indirect addressing modes with an example.	4	2	2	2	2.4.1
15	Describe the role and functions of the Arithmetic and Logic Unit (ALU) and the Memory Unit (MU) in the basic structure of a computer. How do these components interact to perform a simple arithmetic operation?	4	2	2	2	2.4.1
	Part – C (2 x 12 = 24 Marks)					
16	A financial firm is developing a secure data transmission protocol for online transactions. The protocol needs to ensure that all transmitted data is accurate and has not been tampered with. a) Explain how different number systems (binary, decimal, hexadecimal) can be used to encode and transmit financial data securely. Explain the steps of how the data is converted.	12 6	3	1	2	2.4.1
	b) Discuss the binary codes (such as BCD, Excess-3, and Gray) representation in financial data transmission.	6				
	OR					
17	A digital electronics engineer is designing a simple calculator that can handle binary arithmetic operations (addition, subtraction, multiplication, and division). The calculator uses an 8-bit binary number system. Provide the step-by-step process, with calculations, for Arithmetic operations of the binary numbers A → 110101 B → 001111.	12	3	1	2	2.4.1
18	You are designing a simple computer system for a basic educational purpose. The system should be able to perform basic arithmetic operations (addition, subtraction, multiplication, division) and handle memory operations efficiently. Explain the memory addressing for this system. Describe the type of addressing modes you would use and justify your choices.	12	3	2	2	2.4.1
	OR					
19	A system that processes 4-bit information and includes a basic instruction set, such as data transfer and data manipulation instructions, to execute operations. Explain each of these instructions with an example.	12	3	2	2	2.4.1

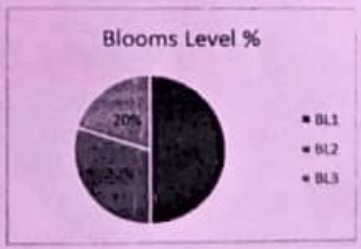
Decimal	58	170
Binary	?	?
Octal	?	?
Hexadecimal	?	?

Decimal	32	50
BCD	?	?
XS-3	?	?
Gray	?	?

*Performance Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator